

Arunabha Majumder

Research Engineer — Wearable Robotics, Exoskeletons, Actuators & Experimental Validation

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Robot Design Lab, Division of Mechatronics
KTH Royal Institute of Technology, Stockholm, Sweden

Personal Letter: Research Engineer in Robot Design and Mechatronics (REMA project)

Dear Members of the Hiring Committee,

I am applying for the position of Research Engineer in robot design and mechatronics with a focus on soft and wearable robotics. I hold an MTech in Mechatronics and a BTech in Mechanical Engineering, and I am completing my PhD in Mechatronics and Robotics at Aalborg University, where I design and test exoskeletons. The REMA project's goal, new exoskeletons that help older adults stay mobile, is exactly the kind of research I want to dedicate the next step of my career to.

Wearable robotics has been the thread through my whole education. My Bachelor's capstone was an upper-limb rehabilitation exoskeleton, which I designed, machined, assembled and tested; it won the Best Major Project Award and a TATA Technologies Innovation Grant. My Master's thesis at CSIR-CMERI focused on soft pneumatic artificial muscles: I fabricated them from raw materials, built the test rig with valves, pressure sensors and DAQ, and developed a neural-network-based gain-scheduled controller in MATLAB/Simulink that reduced tracking error to 0.3–0.78%, published at IEEE CONECCT 2022.

In my PhD project VIEOX, I designed a hybrid-actuated shoulder exoskeleton that combines a compliant parallel spring with a small 6 Nm motor. I carried out the mechanical design in SolidWorks, built the prototypes through 3D printing, and integrated the motor, encoder, force sensors and embedded control. The design reduced energy consumption by more than 25% and received the Best Research Paper Award at IFToMM ISRM 2026. I also implemented an Assist-as-Needed controller that reduced user muscle effort by more than 15%, and on a 5-bar planar parallel robot I developed a force-estimation method using variable-stiffness actuators and the robot Jacobian, validated against a force sensor to below 10% RMS error and used to measure human arm impedance.

Experimental work with people is central to wearable robotics, and I have conducted experiments with 12 participants using EMG and motion capture across several assistance modes and load conditions. I have also supervised Bachelor's students and taught university courses, which matches the student supervision part of this role.

What draws me to Robot Design Lab is the combination the REMA project brings together: soft robotics, sensing, actuation, adaptive control and user-centred design. My experience covers rigid and compliant exoskeleton actuation and soft pneumatic actuators, and I am eager to deepen my work on soft wearable structures, to use multibody simulation such as Simscape Multibody more systematically, and to design with older adults' needs in mind. I would bring a hands-on, iterative approach: building functional prototypes quickly, verifying them experimentally, and turning the results into publications.

I would be glad to contribute to the REMA project and to the research of Robot Design Lab. Thank you for considering my application.

Kind regards,
Arunabha Majumder